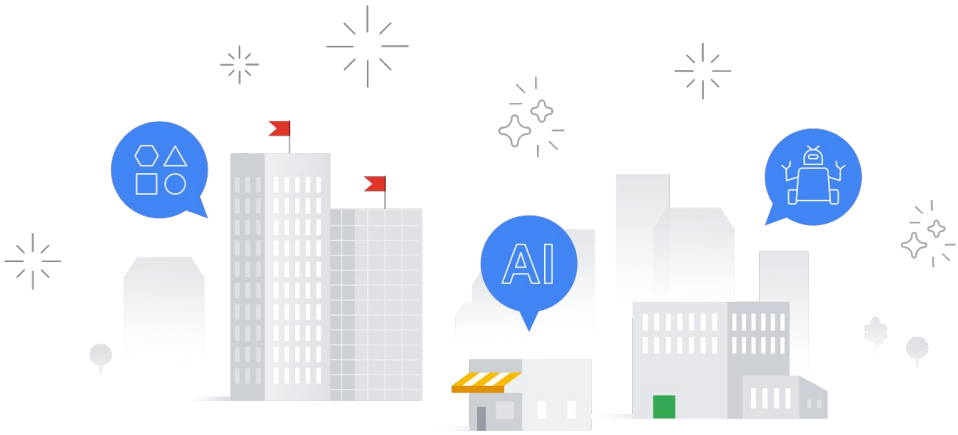
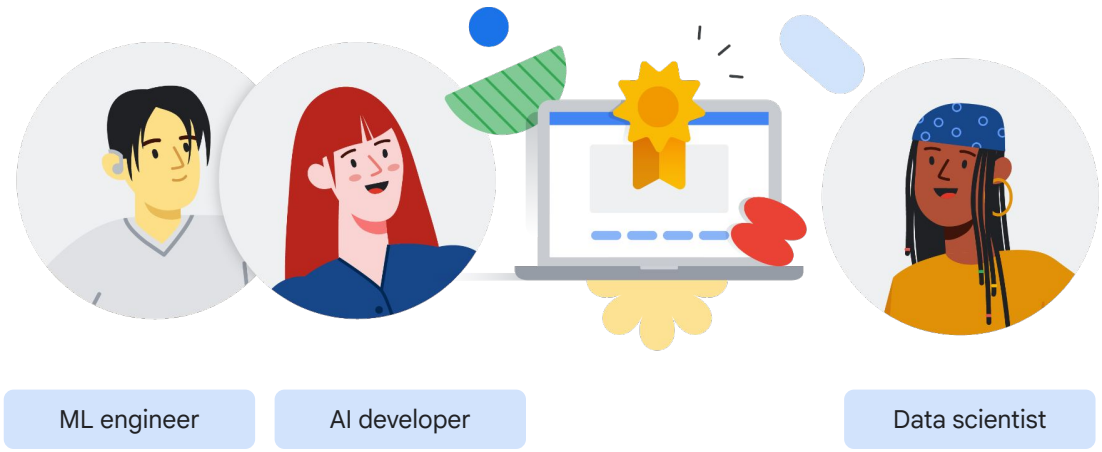


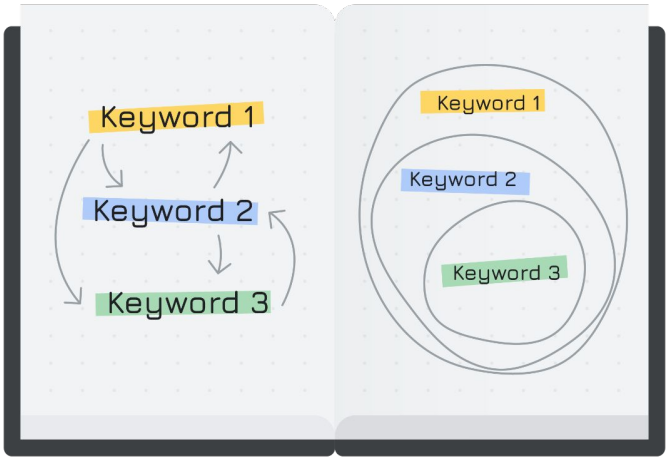
Introduction to AI and Machine Learning on Google Cloud

Congratulations on completing the course, *Introduction to AI and Machine Learning on Google Cloud*!



Regardless of your role—an AI developer, data scientist, ML engineer, or simply a passionate AI and machine learning enthusiast—we trust that this course has provided you with valuable insights to enhance your career.

Keywords in your mind?



Before the final review, let's take a moment to reflect on what you've learned so far. Can you identify the key **AI concepts, technologies, and tools** from the course and organize them in a mind map that shows their connections?

01 AI foundations ☑

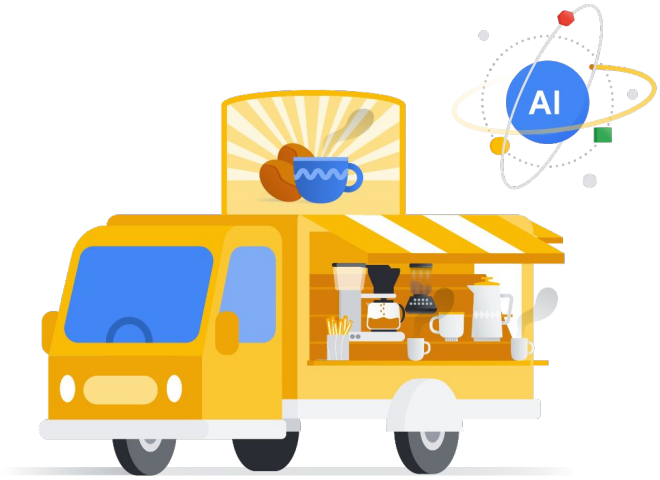
02 Generative AI ☐

03 AI development options ☐

04 AI development workflow ☐

Well, let’s walk through the main concepts. You started from AI foundations.

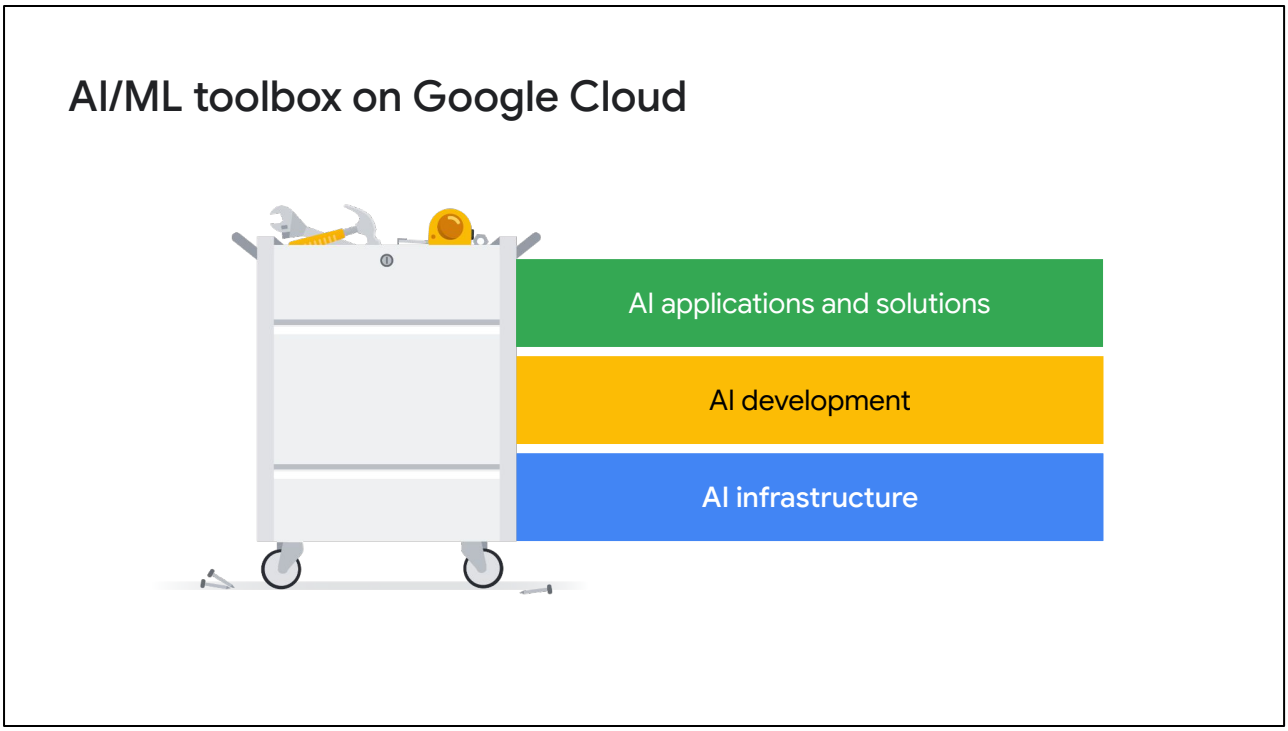
A use case: Coffee on Wheels



- Predictive AI
- Generative AI

Instead of a morning coffee, you kicked off your journey with Coffee on Wheels, a fascinating food truck use case.

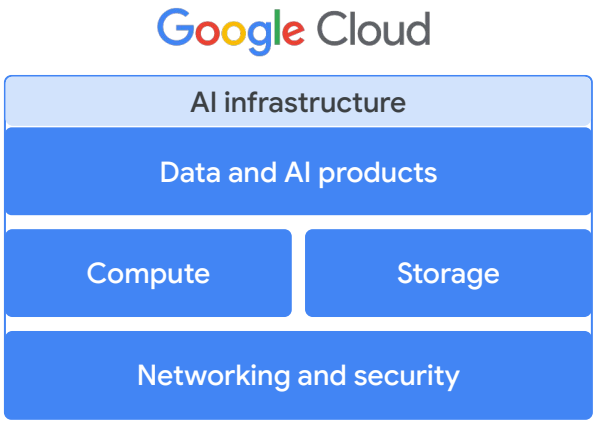
It brilliantly showcased how AI is evolving from a mere efficiency tool to a powerhouse of innovation, tackling both traditional predictive AI and modern generative AI challenges.



To help you achieve your AI vision, you were then introduced to the AI/ML toolbox on Google Cloud, which is structured into three layers:

- AI infrastructure.
- AI development.
- And AI applications and solutions.

AI infrastructure



You started with AI infrastructure in this module, specifically compute and storage.

Data and AI products: Data-to-AI workflow

Ingest and process

- Real-time and batch
- Pub/Sub, Dataflow, Dataproc, and Cloud Data Fusion

Store and analyze

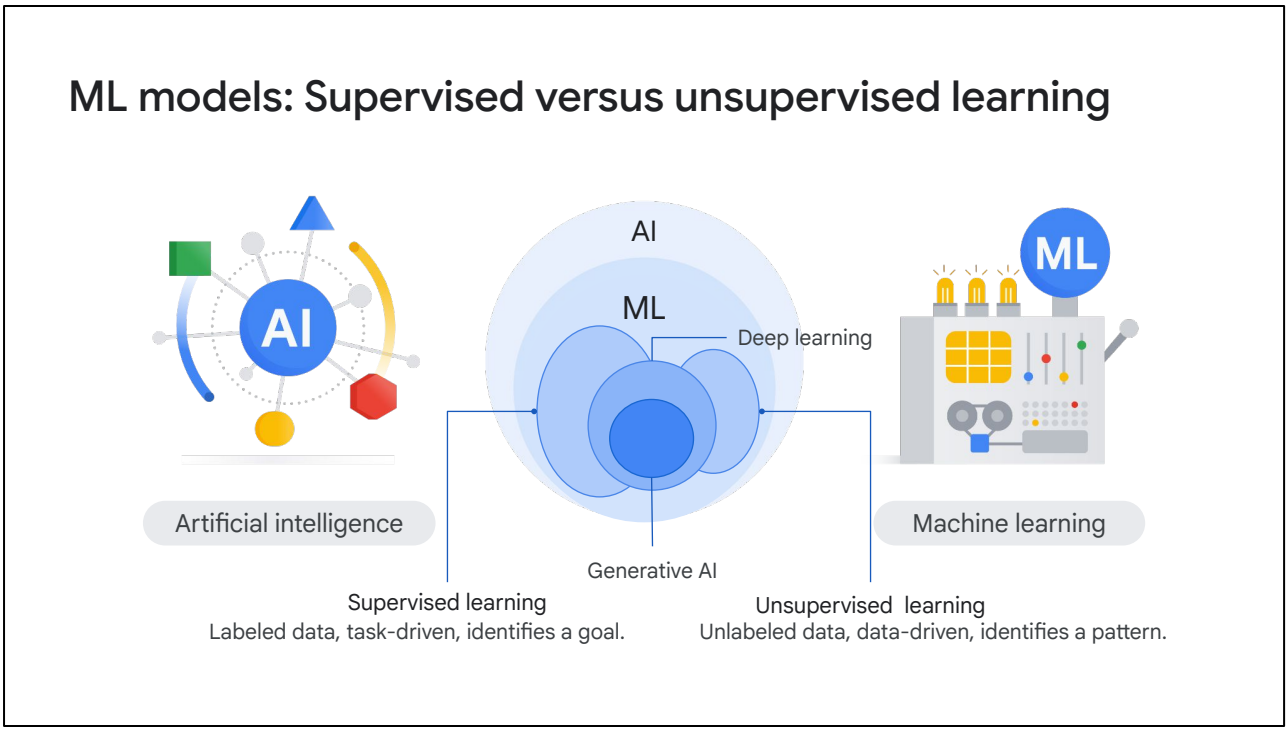
- Store: Cloud Storage
- SQL analysis: BigQuery, AlloyDB, Cloud SQL, and Spanner
- No-SQL: Bigtable and Firestore
- Visualization: Looker

Activate with AI

- AI platform: Vertex AI
- Out-of-box solution to custom builds: Vertex AI Studio, Agent Builder, AutoML, and Notebooks

You even discovered the data and AI products that guide the journey from data to AI, covering:

- Ingestion and processing.
- Storage and analysis.
- And finally, activation with AI.



Before diving into practical applications, you were equipped with AI model basics, focusing on supervised versus unsupervised learning.

Key phases

01 Extract, transform, and load data into BigQuery.

02 Select and preprocess features.

03 Create the model inside BigQuery.

04 Evaluate the performance of the trained model.

05 Use the model to make predictions.

Build an ML model with BigQuery

BigQuery

With these concepts in mind, you rolled up your sleeves and honed in on the module's highlight: BigQuery ML. Here, you explored various models and walked through the steps to build your own ML model.

Hands-on lab:
Predict Visitor Purchases with BigQuery ML

For the fun part, you had a hands-on lab where you applied those steps to build your own ML model using SQL commands.

And, even better, you got to play with your new best friend, Gemini Code Assist, who helped you explain, create, and debug SQL code.

01 AI foundations

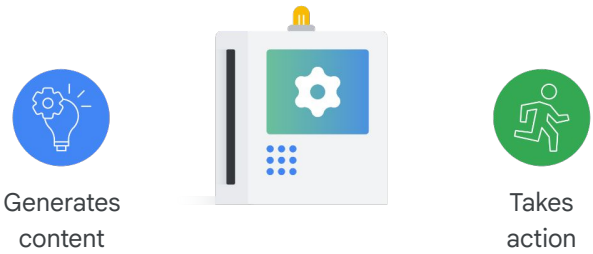
02 Generative AI

03 AI development options

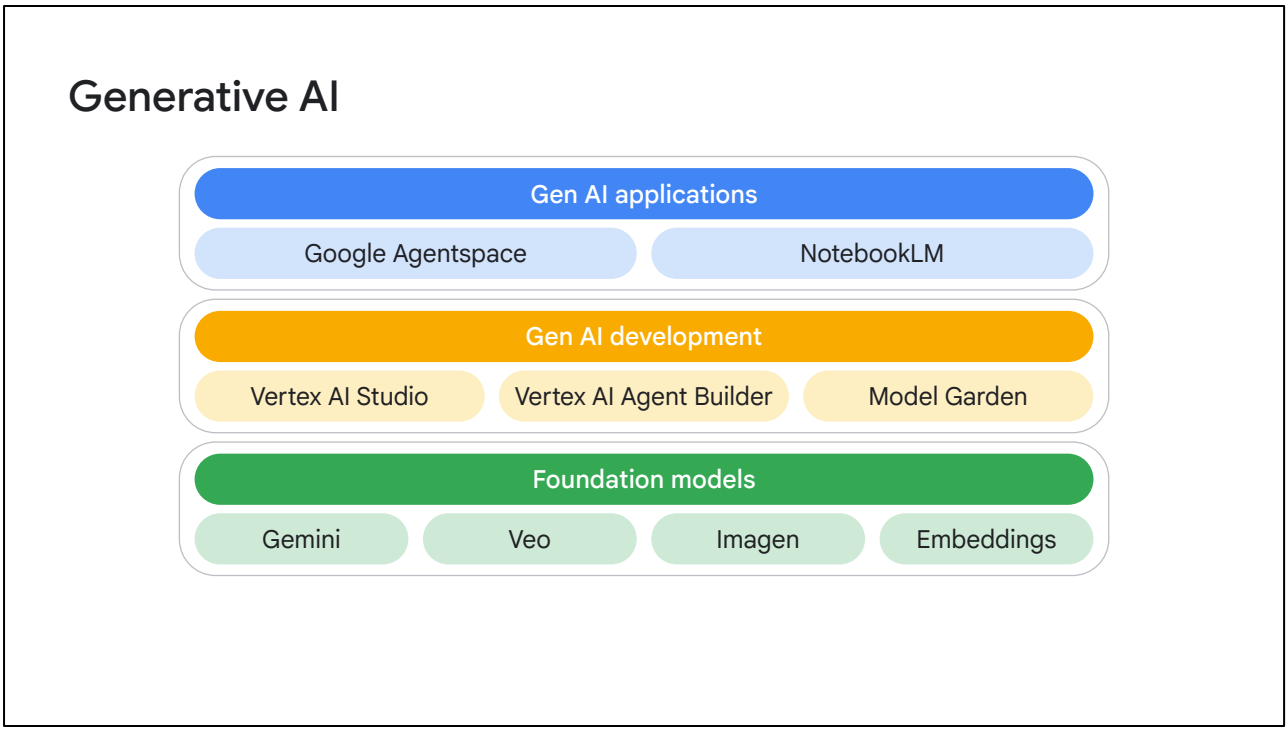
04 AI development workflow

Equipped with AI foundations, you proceeded to generative AI, which is the most recent AI advancement.

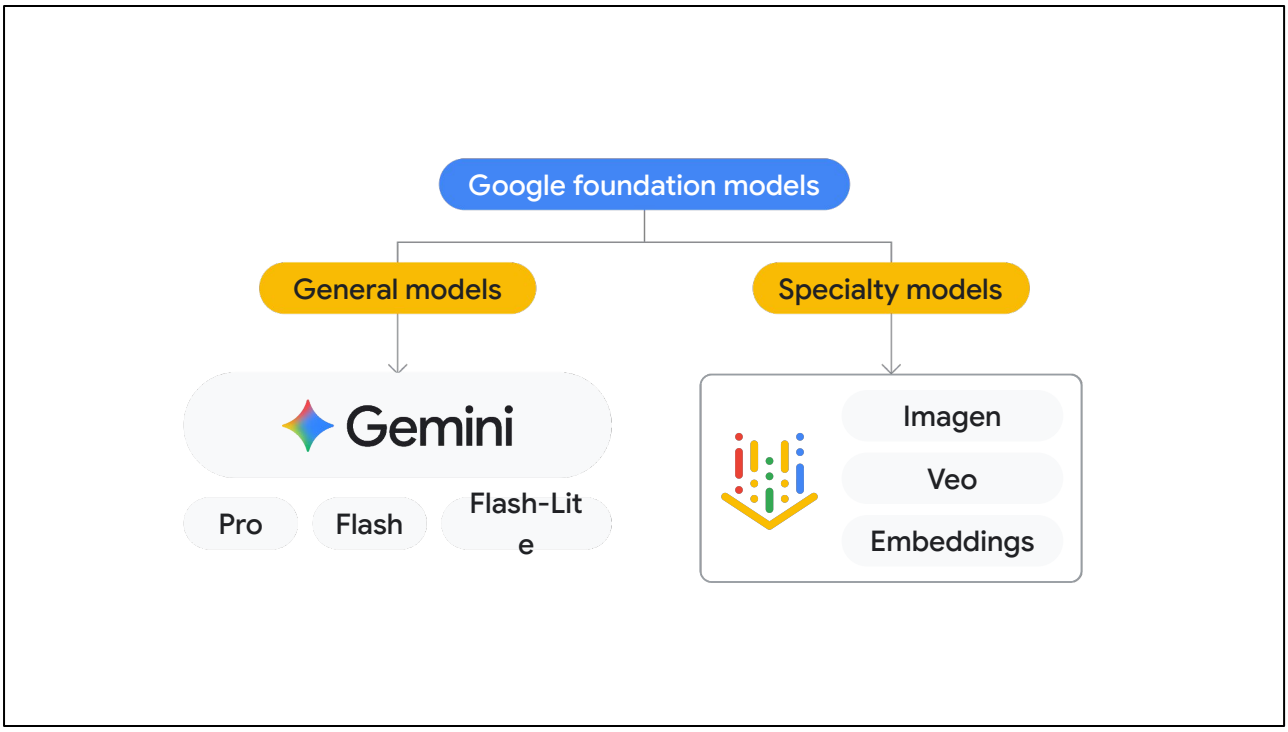
Generative AI



Gen AI not only **creates content** but also **takes action** for you.

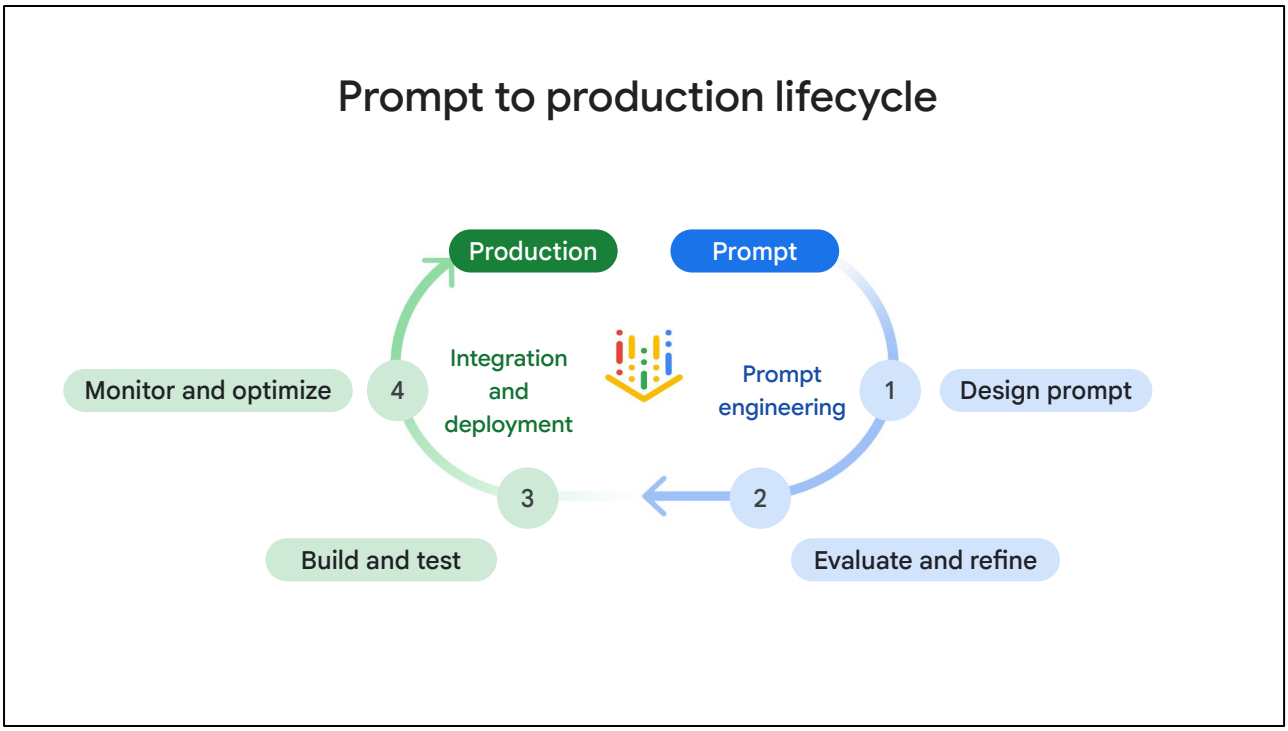


You were first introduced to Google's **three-layered gen AI stack**: foundation models, gen AI development tools, and gen AI applications.



You started with **foundation models**, the backbone of gen AI, which includes Google's general-purpose model (the Gemini family) and specialized models (like Imagen and Veo).

You also learned about **Gemini's multimodal** capabilities, and how to access and tailor these models.



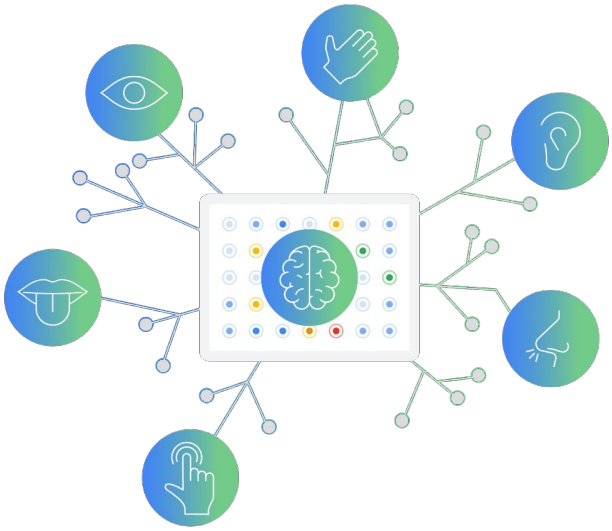
You then explored **Vertex AI Studio** and the prompt to production **lifecycle**.

- You joined Bea, Ann, and Ian from Cymbal Insurance, who quickly **turned an idea into an application** and built a prototype where AI helps with insurance risk analysis. In this process, you learned what components make a good prompt: task, context, and examples.

- After a quick idea-to-app journey, you dived into the first half of the lifecycle: **prompt engineering**. This involved prompt design, evaluation, and refinement, enhanced by Vertex AI Studio's features like prompt templates, model parameter specification (such as temperature, top K, top P), and side-by-side prompt comparison.

- You then proceeded to the second half of the lifecycle: **deployment and model tuning**. Vertex AI Studio streamlines app development with automated code generation and seamless integration with Cloud Run and Cloud Shell. You also learned about grounding and RAG for improved model accuracy, alongside various tuning techniques, including prompt design, parameter-efficient tuning, and full fine-tuning.

Three components of an AI agent



The model (brain)

- AI foundation models
- Reasoning center

The tools (hands, feet, and senses)

- The connectors
- Interactions with the outside world

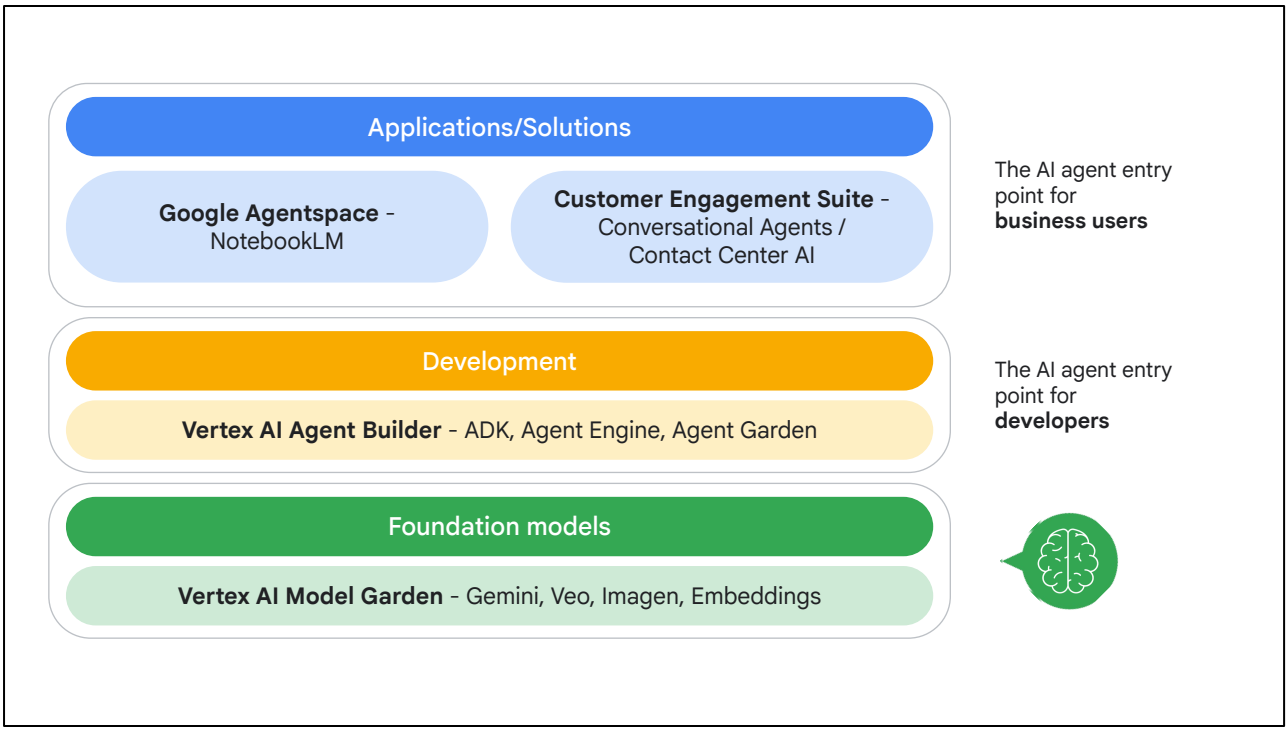
The orchestration layer (nervous system)

- Process and workflow
- Communication and coordination

Next, the journey got more exciting, as you were introduced to **AI agents** as the next evolution of gen AI, moving beyond chatbots and enabling AI to take action, automate workflows, and make decisions.

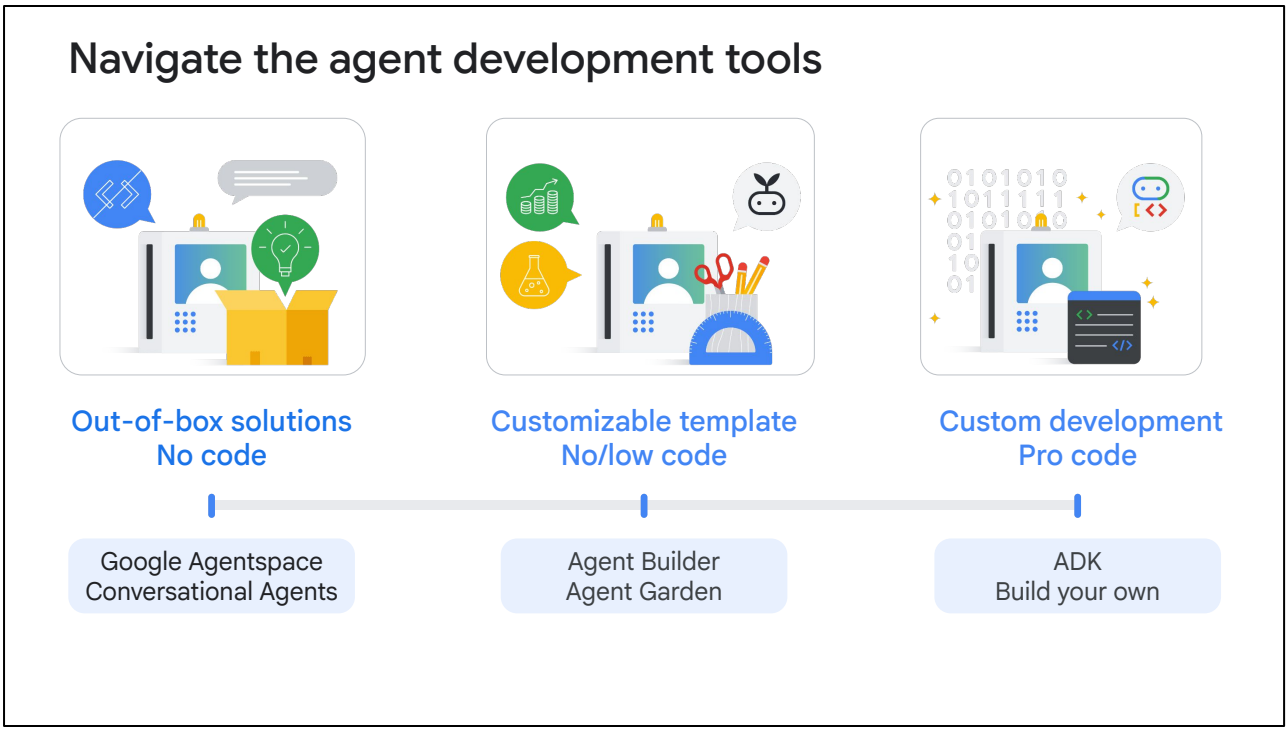
This defines an AI agent by its three **core components**:

- The model (or brain),
- The tools (like hands and feet to connect with the external world)
- And the orchestration layer (or the nervous system)



With the idea of AI agents and agentic AI, the last lesson provided a practical guide on **building AI agents on Google Cloud**.

It highlighted the comprehensive suite of tools offered across Google's generative AI architecture, including Vertex AI Model Garden for accessing and fine-tuning foundation models, Vertex AI Agent Builder for developers to construct AI agents, and both Google Agentspace and NotebookLM for end-users to develop no-code agents.



This lesson also offered a decision tree to help navigate these tools.

Lab:
Get Started with Vertex AI Studio

01 Create applications from prompts.

02 Design effective prompts.

03 Engineer and manage prompts.

04 Use multimodal prompts.

You got time to play with Vertex AI studio and build your own applications.

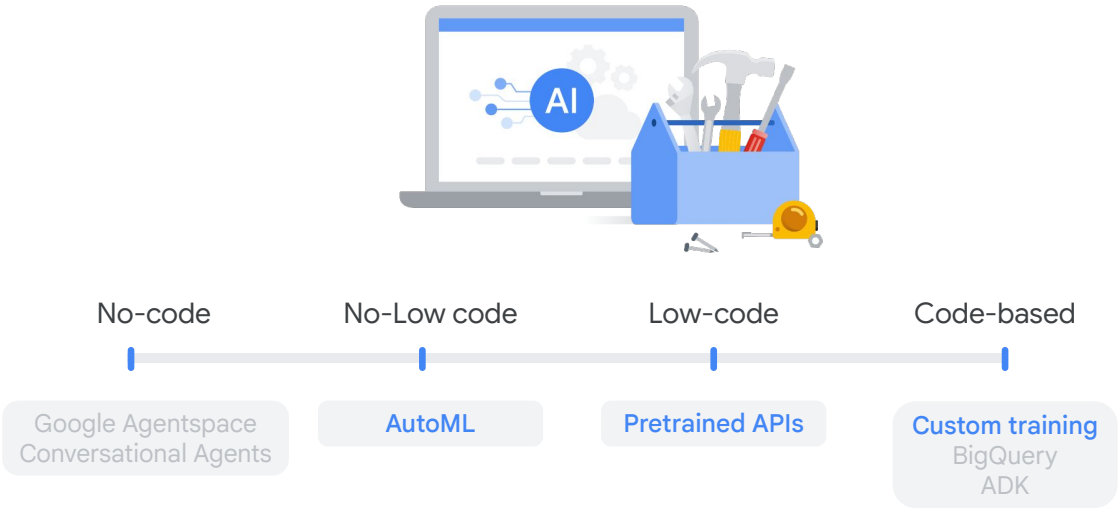
01 AI foundations ☑

02 Generative AI ☑

03 AI development options ☑

04 AI development workflow ☐

In the next module, you explored the question: What options do you have for building an AI project or an ML model? Let's take a quick look.



Google Cloud offers a wealth of options to suit your needs, whether you're a business user, data scientist, developer, or ML engineer.

These range from no-code, out-of-the-box solutions to low-code tools and even code-based DIY approaches.

You've been introduced to Vertex AI, Google's unified AI platform, which serves as your ultimate workspace for exploring these options and building end-to-end machine learning projects.
You've specifically focused on:

- **AutoML:** A fantastic low- or no-code tool within Vertex AI that automates ML development from data preparation to model training and serving, all using your own data.
- **Pre-trained APIs:** Ready-made solutions that leverage powerful pre-trained machine learning models, completely eliminating the need for any training data.
- **Custom training:** This empowers you to manually code ML projects using versatile tools such as Python, JAX, and Vertex AI Workbench.

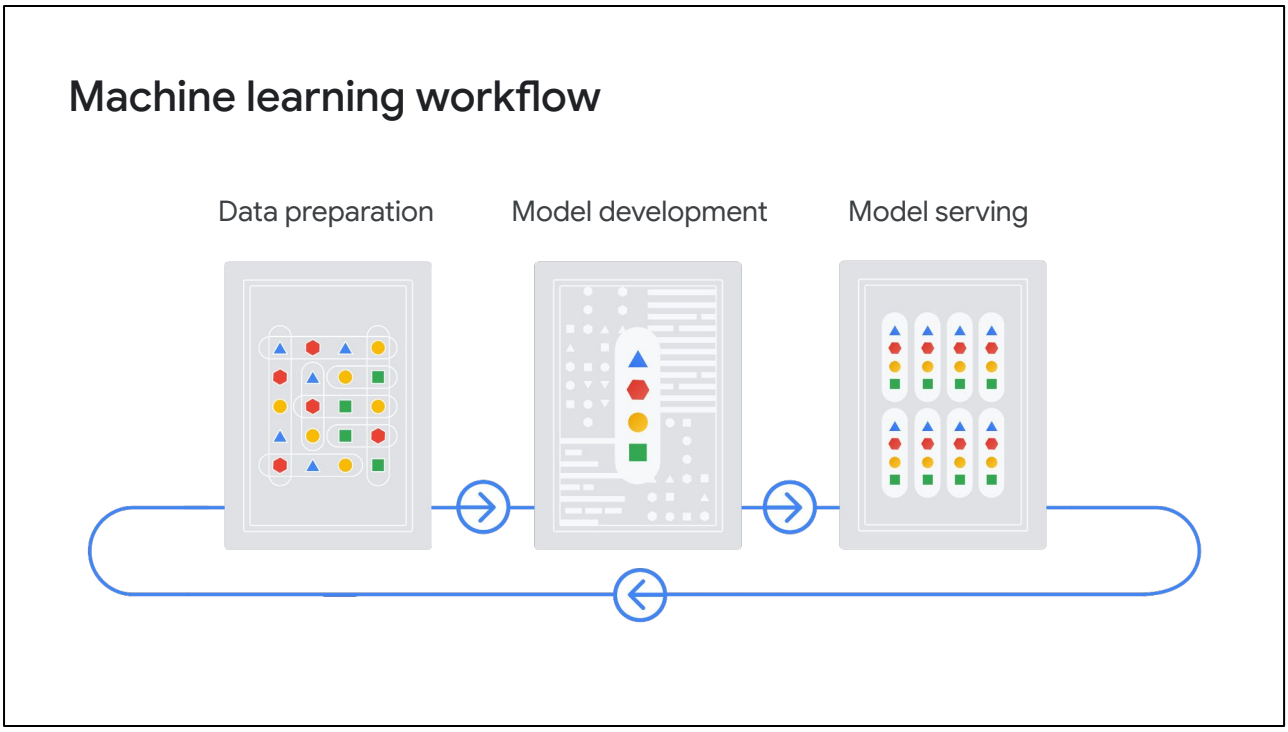
01 AI foundations

02 Generative AI

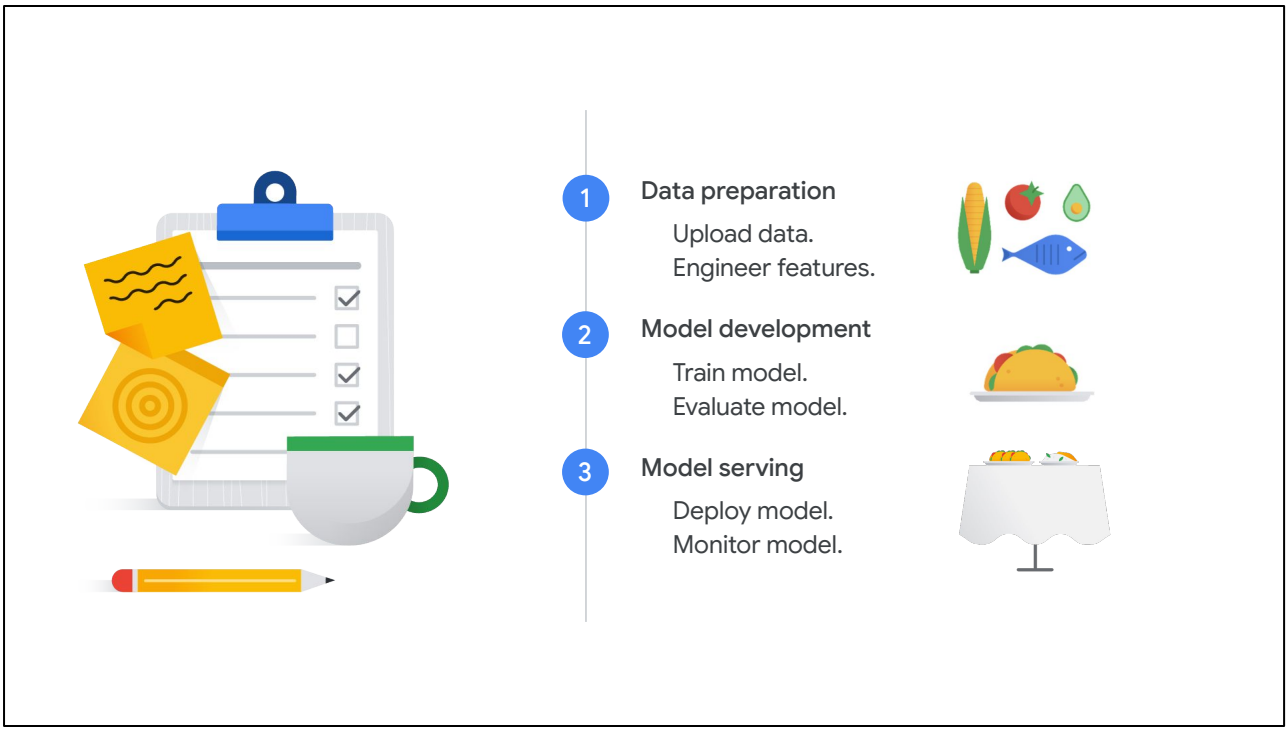
03 AI development options

04 AI development workflow

In the last module, you dived deep and explored the AI development workflow.



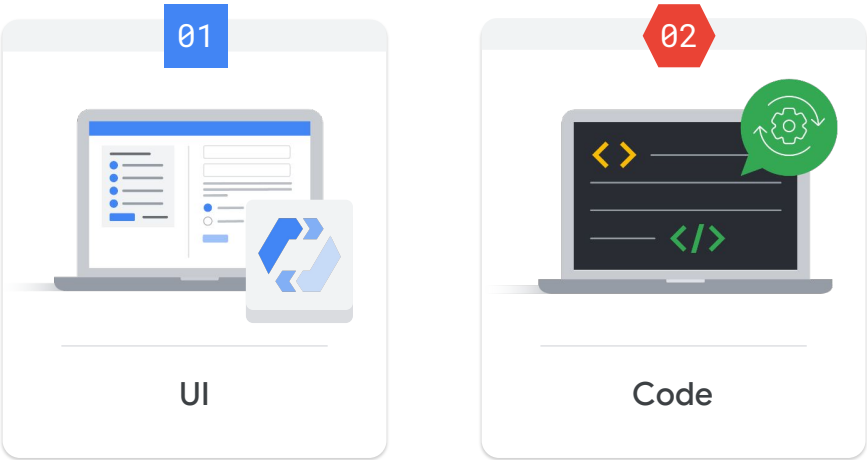
In this module, you learned about the three main stages of the machine learning workflow with the help of the restaurant analogy.



In stage one, data preparation, you uploaded data and applied feature engineering. This translated to gathering our ingredients and then chopping and prepping them in the kitchen.

In stage two, model development, the model was trained and evaluated. This is where you experimented with the recipes and tasted the meal to ensure that it turned out as expected.

And in the final stage, model serving, the model was deployed and monitored. This translates to serving the meal to customers and adjusting the menu as more people tried and reviewed the dish.

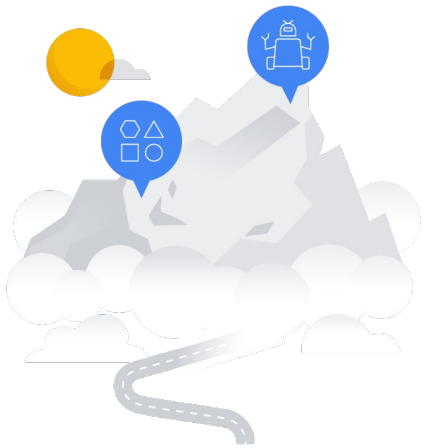


There are two ways to build a machine learning model from end to end:

- One is through a **user interface**, like you practiced in the AutoML lab.
- The other is with **code**, which you were shown using prebuilt SDKs with Vertex AI Pipelines. The latter helps you automate the ML pipeline to achieve continuous integration, training, and delivery.

Thanks for joining us in the kitchen, where you explored the full recipe from prepping data to serving models.

We can't wait to see you build your own ML models with Google Cloud!

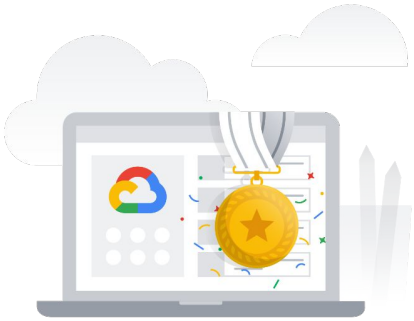


Let this course kickstart your journey into AI and machine learning.

Remember what we suggested at the beginning of the course: apply what you've learned to your own work. This is the best way to develop your skills as an AI practitioner.

cloud.google.com/training/machinelearning-ai

For more training with machine learning and AI, please explore the options available at cloud.google.com/training/machinelearning-ai.



Google Cloud certification

cloud.google.com/certifications

If you are interested in validating your expertise and showcasing your ability to transform businesses with Google Cloud technology, you may want to consider pursuing a Google Cloud certification. You can learn more about Google Cloud's certification offerings at: cloud.google.com/certifications.

Thanks for completing this course.

We'll see you next time!